

F, N = number of inputs/steps per training sequence, number of layers (not inc. input layer)
for each epoch:

for each x_sequence, y in training set:

for each layer L in network from 1 to N inclusive:

$a^{L,0}$ = an appropriately sized column vector of zeroes

for each step S from 1 to F inclusive:

$a^{0,S} = x_sequence[S]$ # or possibly $[S-1]$

for each layer L in network:

$dot^{L,S} = w1^L \cdot a^{L-1,S} + ws^L \cdot a^{L,S-1} + b^L$

$a^{L,S} = A(\text{_____})$

$\Delta^{N,F} = A'(dot^{N,F}) \otimes (\text{_____})$

for each step S (counting down from F-1):

$\Delta^{N,S} = A'(dot^{N,S}) \otimes (\text{_____})^T \cdot \Delta^{N,S+1}$

for each layer L in network (counting down from N-1):

$\Delta^{L,F} = A'(dot^{L,F}) \otimes (\text{_____})^T \cdot \text{_____}$

for each step S (counting down from F-1):

for each layer L in network (counting down from N-1):

$\Delta^{L,S} = \text{_____} + \text{_____}$

for each layer L in network:

$b^L = b^L + \lambda \cdot \text{sum}(\Delta^{L,S} \text{ for all } S)$

$w1^L = w1^L + \lambda \cdot \text{sum}(\Delta^{L,S} \cdot (\text{_____})^T \text{ for all } S)$

$ws^L = ws^L + \lambda \cdot \text{sum}(\text{_____} \cdot (\text{_____})^T \text{ for all } S \text{ except the first})$